
The Feature Networks of High Medieval Writing

Celis Tittse^{*1,2}

¹Utrecht University – Netherlands

²KU Leuven Faculty of Theology – Belgium

Abstract

The long twelfth century (1075–1225) is characterised by a relative explosion of manuscript production and transformations in script (Kwakkel and Thomson 2018, 25). Rather than constituting a linear transition from Carolingian to Gothic book culture, these changes emerge without clear cohesion within the palaeographical features across time and space (Kwakkel 2012). This heterogeneity provides a productive basis for applying a historical network-based analysis to identify similarities between scribal communities.

This paper treats palaeographical features themselves as historical sources for network construction (Salamon 2017, 10; Johnston and Van Dussen 2015, 25). Using Kwakkel’s enriched dataset from the *Catalogue des Manuscrits Datés* (351 dated manuscripts; 135 with reliable localisation), we model scribal relationships through the co-occurrence and circulation of letterform features (angularity, minims, and biting) (Derolez 2003, 51–56): two complementary network experiments are presented.

First, feature co-occurrence networks are constructed within successive half-century time slices and the full dataset. Pairwise associations are evaluated using Fisher’s exact test ($p < 0.05$) and Cramér’s V ($f > 0.3$), allowing us to identify statistically significant feature communities while explicitly accounting for sparse and uneven source distributions. This analysis confirms known associations, such as the coupling of Gothic minims with angularity (Figure 1).

Second, these found *feature communities* are projected on a geo-temporal manuscript network. Manuscripts sharing identical feature combinations are linked to their closest temporal neighbours within 20-year intervals, producing temporally constrained, spatially networks of scribal practice. These networks reveal specific localised experimentation, phased diffusion, and transitional zones that are difficult to capture through traditional approaches (Figure 2, 3) (Bischoff 1989).

By modelling palaeographical features as network-generating sources, this study shows how statistically grounded feature communities can highlight the cohesion of High Medieval scribal practices. Future work will extend this approach through automated letterform extraction on a larger corpus, enabling more fine-grained, computational analyses of medieval scribal networks (Vlachou-Efstathiou et al. 2024, 13).

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^{*}Speaker

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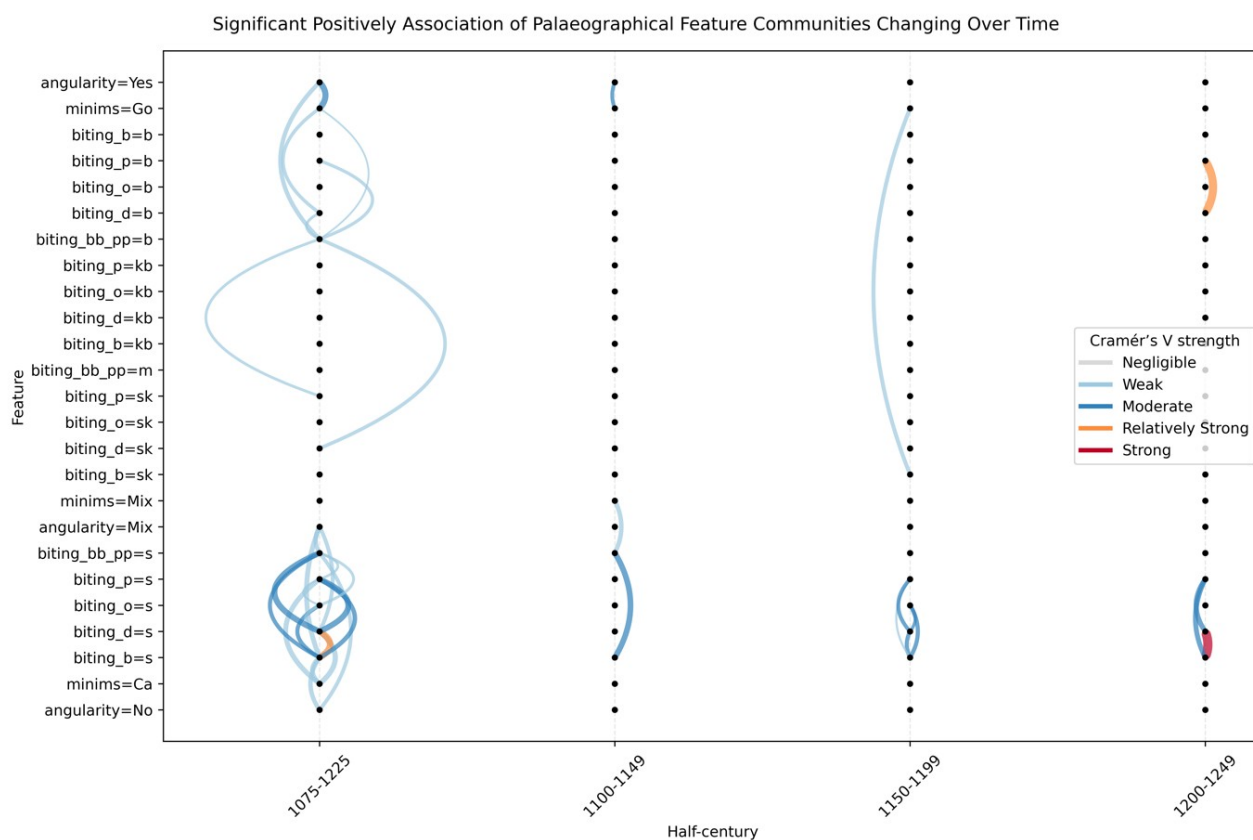


Figure 1: Statistically significant positive association between palaeographical features from Kwakkel's dataset.

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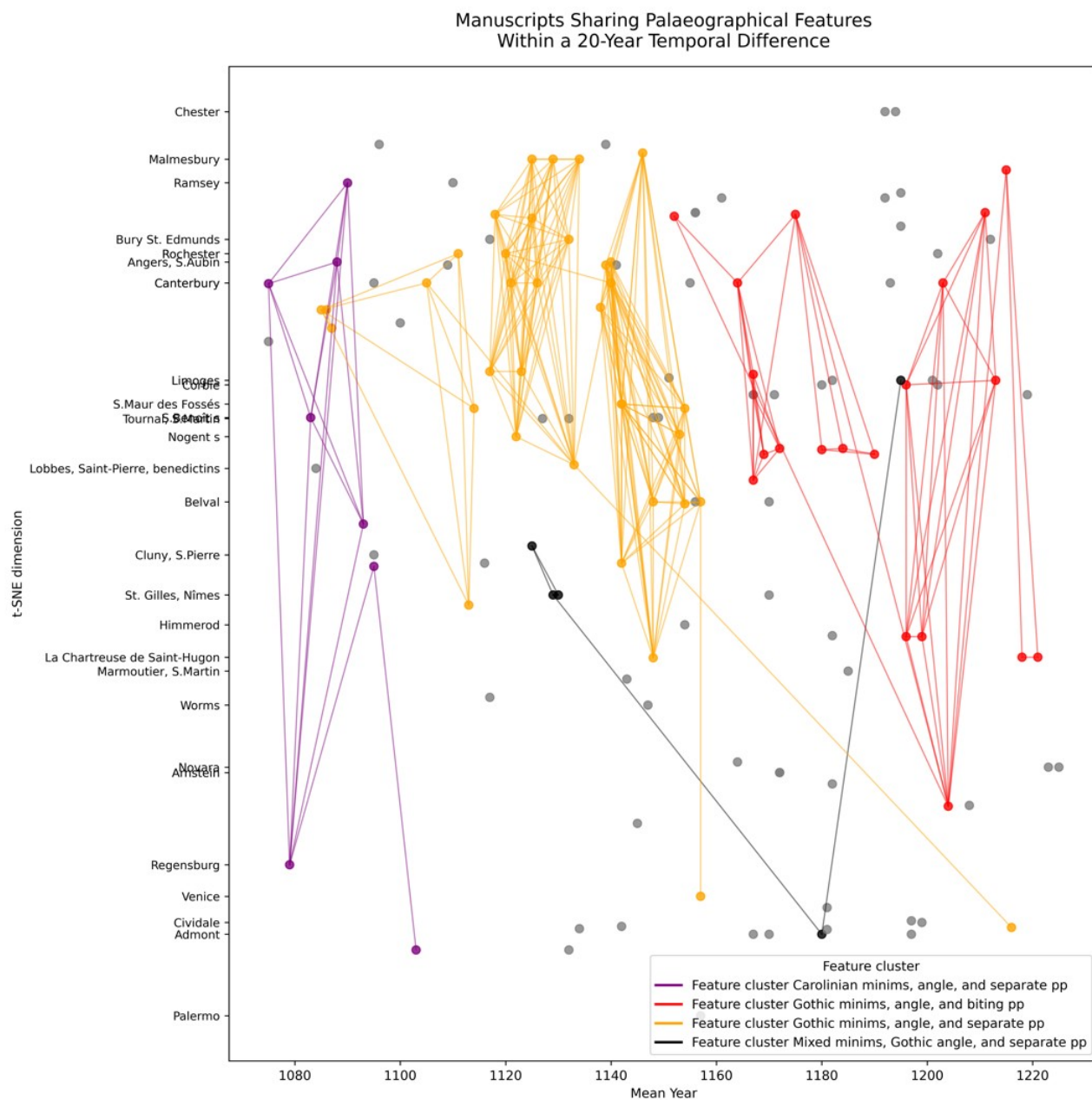


Figure 2: Geo-temporal visualisation of manuscripts with the same palaeographical features within the same 20-year timespan, clustered by 4 most frequent similar features.

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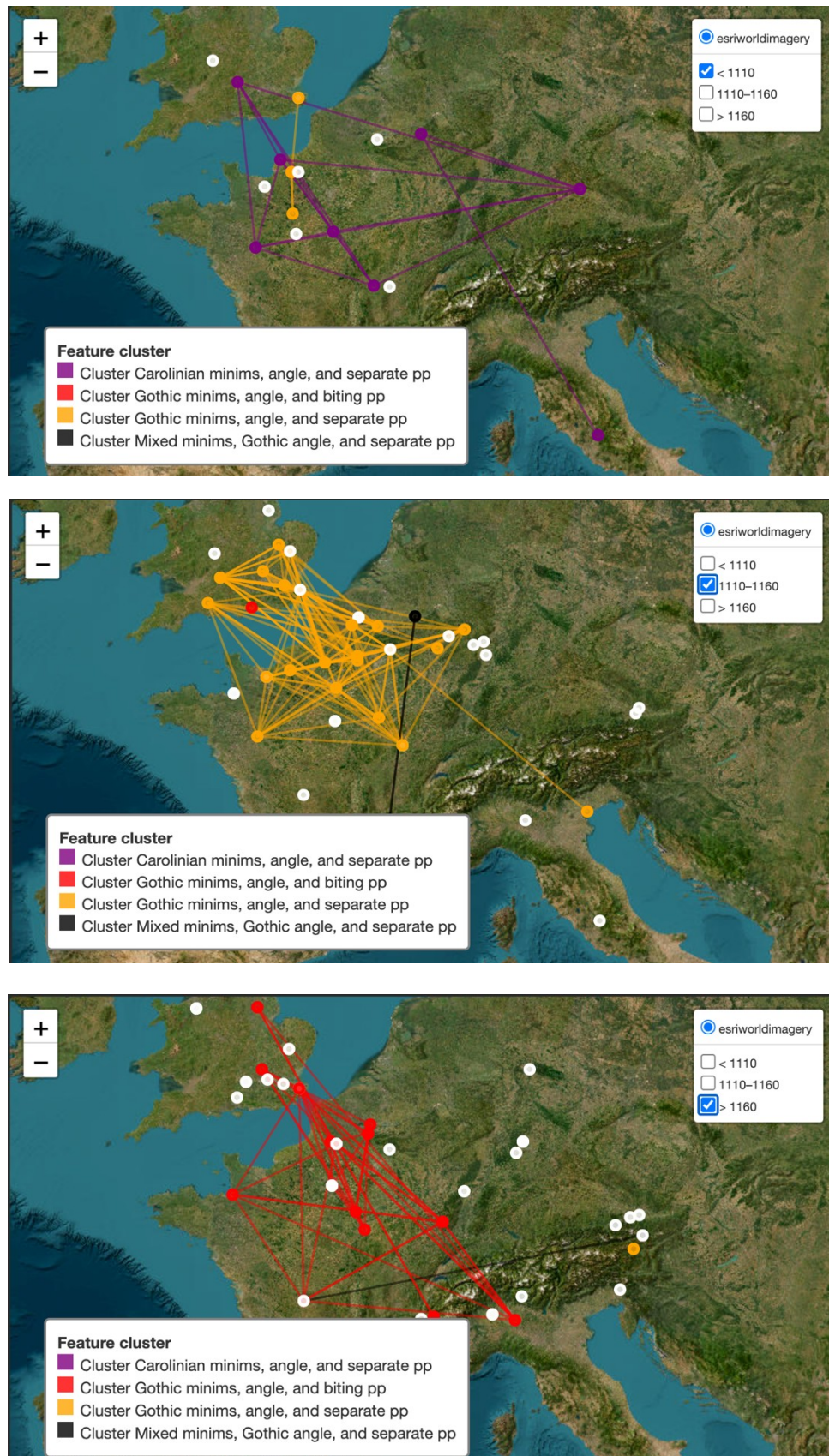


Figure 3: The same data as visualised in Figure 2, but sliced into 3 timeframes (1075-1110, 1110-1160, and 1160-1225) on a map.